THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The decimal value of 10110_2 is: 1

(a) 24 **(b)** 28 **(c)** 22 **(d)** 20

2. The binary form of 45 is: 1

(a) 110001 **(b)** 101100 **(c)** 101101 **(d)** 101110

3. XLIV + LXVII equals: 1

(a) CI **(b)** CXXI **(c)** CXI **(d)** CXII

4. The largest 3-digit number in base 2, written in decimal, is: 1

(a) 7 **(b)** 8 **(c)** 5 **(d)** 3

5. The number of digits in the binary form of 20 is: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 7

6. $1011_2 + 110_2$ equals: 1

(a) 10001_2 **(b)** 10010_2 **(c)** 10000_2 **(d)** 10011_2

7. In 2403_5, the value (in decimal) contributed by the digit 4 in position 2 from the left is: 1

(a) 29 **(b)** 100 **(c)** 20 **(d)** 500

8. The decimal value of 11011_2 is: 1

(a) 33 **(b)** 25 **(c)** 29 **(d)** 27

9. The binary form of 54 is: 1

(a) 110101 **(b)** 110111 **(c)** 110110 **(d)** 011011

10. LXXVIII + CXXXIV equals: 1

(a) CCXIII **(b)** CCII **(c)** CCXII **(d)** CCXXII

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Convert 110101_2 to decimal. 2

12. Write CXXXIX + CCLXII as a Roman numeral. 2

13. Subtract $11010_2 - 1001_2$ and give the answer in binary. 2

14. Convert 20121_3 to decimal. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Convert 101101_2 to base 8 by first converting to decimal. 3

16. Multiply $101_2 \times 11_2$ and give the answer in binary. 3

17. Write 1994 and 449 in Roman numerals and find their difference in Roman numerals. 3

OR

Convert 1432_5 to base 8 by first converting to decimal. 3

18. Multiply $110_2 \times 101_2$ and give the answer in binary. 3

OR

Write 2024 and 1999 in Roman numerals and find their difference in Roman numerals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the sum of 1101_2, 1001_2 and 110_2. Verify the result in decimal. 5

OR

Write the smallest and the largest 5-digit numbers in base 2 and find their decimal values. How many 5-digit numbers are there in base 2? 5

20. Find the sum of 10001_2, 1011_2 and 101_2. Verify the result in decimal. 5

OR

Write the smallest and the largest 4-digit numbers in base 3 and find their decimal values. How many 4-digit numbers are there in base 3? 5

END OF QUESTION PAPER — Total: 40 Marks

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